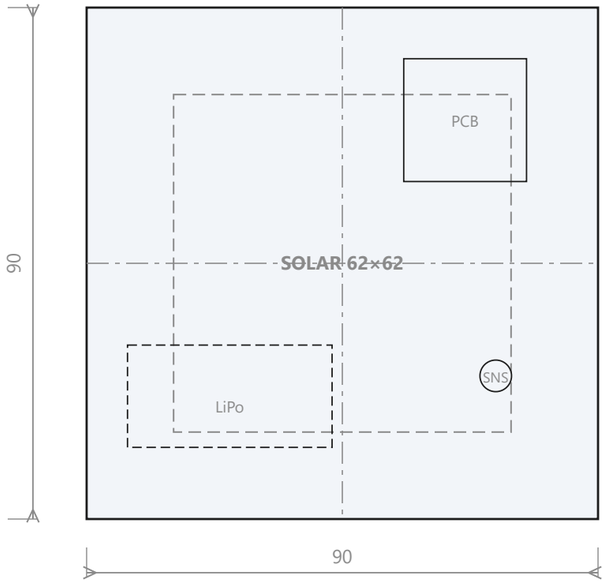
S · MITE
GENERAL ARRANGEMENT — SWARM UNIT

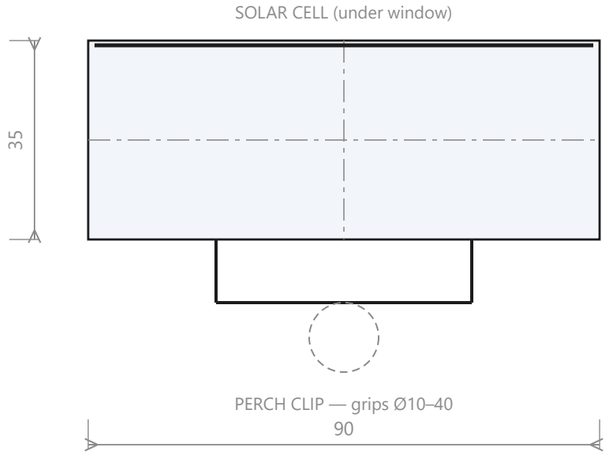
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FOR REVIEW

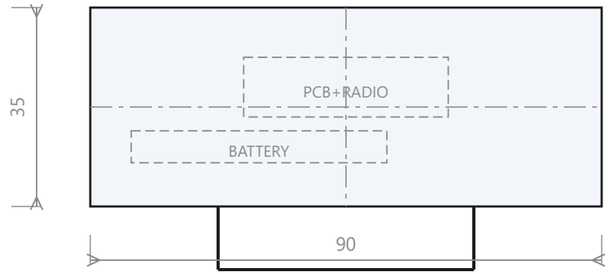
TOP VIEW



FRONT VIEW



SIDE VIEW



WHAT THIS NODE IS

— One member of the fire-watch swarm — the swarm IS the sensor, not the individual bee.
— Minimal on purpose: solar top, LoRa radio, temp/RH/VOC sensing, perch clip.
— Default state = perched and reporting. Flight is a later role, not the power budget.
— Same signed mesh identity as every ArcticChain node.

SWARM ROLES (ASSIGNED BY MESH AI)

ROLE	PAYLOAD	JOB
Sensor-bee	temp/RH + VOC	sniff anomalies — primary detector
Eye-bee	thermal/visual (rev-C)	converge on fixes to confirm
Relay-bee	LoRa only, high perch	extend range over ridges
Charger-bee	dock contacts	perch at Anchor, top off, rebroadcast

GEN-1 VS ROADMAP

STAGE	WHAT	GATE
Gen-1 (now)	~90 mm puck, solar+LoRa+gas, perch clip	orderable parts only
Gen-2	smaller, printed antenna, lower cost	volume + field data
Gen-3 (R&D)	sub-bee perched motes, degradable	transient-electronics — see D SEED

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SHEET **1/7**
DWG NO **AF-S-101**
UNITS **mm**

PARTS LIST — GEN-1 (ALL ORDERABLE IN CANADA)

#	PART	SUPPLIER	APPROX CAD	QTY
1	ESP32-S3-DevKitC-1 / S3-WROOM carrier	PiShop / DigiKey	$8–15	1
2	EBYTE E22-900M30S SX1262 LoRa 915 MHz	DFRobot / AliExpress	$15–25	1
3	Sensirion ENS160 VOC/AIQ (I2C)	DigiKey / Mouser	$6–10	1
4	Sensirion SHT41 temp/RH (I2C)	DigiKey / Mouser	$4–7	1
5	60×60 mm ~1 W 5 V poly solar cell	Amazon.ca / supplier	$5–10	1
6	503450 LiPo 2000 mAh + JST-PH	Amazon.ca / RC supplier	$12–18	1
7	CN3791 MPPT solar charge module	AliExpress / Amazon.ca	$3–6	1
8	Custom carrier PCB ~38×38 mm (JLCPCB)	JLCPCB / PCBWay	$15–30	1
9	Printed enclosure set + spring strip	Print farm	$5–10	1

POWER BUDGET (ESTIMATES — VERIFY ON BENCH)

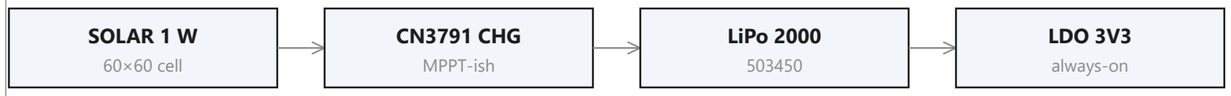
STATE	CURRENT	DUTY	ENERGY/DAY
Deep sleep (ESP32)	~10 µA	~95%	~0.003 Wh
Sense + process	~80 mA @3.3 V	~4%	~0.013 Wh
LoRa TX burst	~120 mA @3.3 V	~1%	~0.010 Wh
RX listen windows	~15 mA	~5%	~0.006 Wh
TOTAL			≈0.03 Wh/day

2000 mAh LiPo ≈ 7.4 Wh → ~200 days theoretical; winter + self-discharge → plan 60–90 day autonomy.

CARRIER PCB PIN MAP

NET	ESP32-S3 GPIO	FUNCTION
LORA_SCK/MISO/MOSI	12 / 13 / 11	SX1262 SPI
LORA_NSS/RST/BUSY	10 / 9 / 8	chip select · reset · DIO1 IRQ
I2C_SDA/SCL	1 / 2	ENS160 (0x53) + SHT41 (0x44)
BATT_ADC	4	battery voltage sense
POGO_CHG	3	dock charge detect
SOLAR_OK	5	charge status

SYSTEM BLOCK



ESP32-S3 + SX1262 + ENS160 on 38×38 carrier; sensors on I2C; radio on private SPI.

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UNITS	**mm**

ASSEMBLY SEQUENCE

1. Print shell, lid, battery bay, clip — ASA/PC-ABS shell, PETG clip.
2. Populate carrier PCB: ESP32-S3, SX1262, ENS160, SHT41, charge circuit.
3. Seat battery, connect JST; verify charge path (bench supply or solar sim).
4. Close lid — gasket bead, solar under window film.
5. Attach perch clip; bench-test join to test mesh.
6. Field trial: perch ×3 units on fence line, log 7 days.

FIRMWARE

ITEM	DETAIL
Runtime	ESP-IDF (C) gen-1; Rust port when mesh-net envelope lands
Envelope	same signed format as fleet — MUST match mesh-net
Duty cycle	deep-sleep 95% · sample 5 min · TX on anomaly/schedule
Data	temp/RH/VOC + battery + RSSI + role → <200 B frame
OTA	signed updates over mesh (rev-C); USB/SD bench flash

ROLE ASSIGNMENT (SKETCH)

— All Miters broadcast hello + battery + capability every N hours.
— Anchor/Brick AI computes coverage gaps → assigns roles via signed cmd.
— Sensor-bees concentrate near fuel breaks/ridgelines in fire season.
— Charger-bees migrate to Anchor docks when battery <20%.
— Roles are soft state — units self-organize if coordinator is gone.

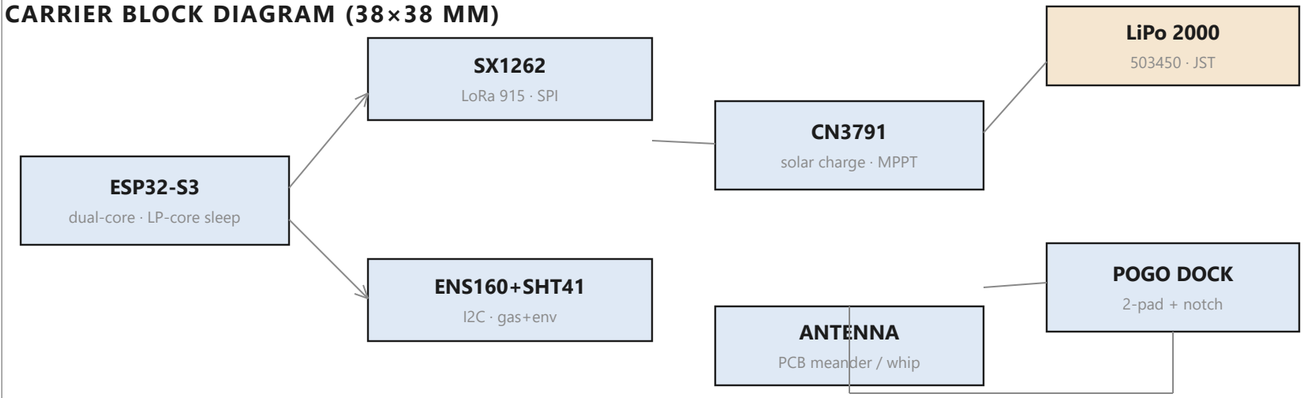
ACCEPTANCE / TEST CRITERIA

TEST	PASS CONDITION
Mesh range	signed observations A→B via Brick/Anchor ≥3 km LOS
Autonomy	≥60 days at 1 sample/10 min + hourly beacon
Cold	operates -30 °C; solar resumes charge at -10 °C
Perch	clip holds Ø25 mm branch through wind-shake test
False positive	<1 smoke alarm/week/unit in dust/inversion

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CARRIER BLOCK DIAGRAM (38×38 MM)



ESP32-S3 = brain · SX1262 = voice · sensors on shared I2C · charge path independent of compute

SLEEP DOMAINS (WHO STAYS ON)

DOMAIN	STATE IN DEEP SLEEP	WAKE SOURCE
ESP32-S3	ULP RISC-V only	RTC timer / GPIO
SX1262	sleep (~µA) + CAD armed	DIO1 / host poll
ENS160+SHT41	power-gated OFF	ESP32 rail control
CN3791	always (harvesting)	— hardware
RTC	always	32 kHz crystal

RADIO & TIMING

PARAM	VALUE
LoRa SF / BW	SF7 / 125 kHz — short range bursts
TX power	+20 dBm (E22-900M30S)
Antenna	PCB meander 915 MHz or ¼-wave whip
Sample	every 5–10 min (temp/RH/VOC)
Beacon	every 60 min signed hello
Anomaly TX	immediate, then 5-min confirm
CAD listen	2 × 200 ms windows/hour

WAKE-UP CHAIN

— Solar voltage sense → wake when dawn charging starts.
— RTC alarm → periodic sense window.
— SX1262 CAD detect → swarm traffic wakes host.
— Dock pogo detect → charge mode + beacon boost.
— All wake sources logged — swarm energy is mesh data.

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UNITS **mm**

VARIANT MATRIX

VARIANT	PAYLOAD DELTA	EXTRA PARTS	COST / DETECTION ROLE
SENSOR-BEE	ENS160 + SHT41 (base)	— none —	+$0 · primary gas/temp anomaly
EYE-BEE	thermal IR array (rev-C)	AMG8833 or MLX90640 + lens	+$25–45 · confirm fixes visually
RELAY-BEE	tall antenna, no sensors	915 MHz whip 8 dBi	+$8–15 · extend range over ridges
CHARGER-BEE	dock pads + charge LED	2 pogo + indicator	+$3–5 · perch at Anchor, rebroadcast

PHYSICAL DIFFERENCES

VARIANT	WHAT CHANGES PHYSICALLY
SENSOR-BEE	base puck — solar window + gas vent membrane
EYE-BEE	domed IR window replaces solar center (rev-C)
RELAY-BEE	extended top mast, whip antenna, minimal face
CHARGER-BEE	bottom pogo pads exposed, dock-align notch

ROLE SELECTION LOGIC

— All variants share: ESP32-S3 + SX1262 + solar + LiPo + perch clip + mesh identity.
— Coordinator (Anchor/Brick) assigns role via signed command — soft state, self-heals.
— Fire season: 70% sensor-bees at fuel breaks, 20% relay on ridgelines, 10% chargers.
— Non-fire season: roles rotate to flood/snowpack/beetle surveillance automatically.
— Eye-bees are rev-C — thermal array needs carrier respin; sensor-bees fly first.

FIELD HIRING (HOW ROLES GET FILLED)

— Swarm joins mesh → coordinator reads capability flags.
— Gap analysis: coverage map × season threat model → role demand.
— Nearest capable unit gets signed role assignment.
— Unit confirms, adopts payload profile, reports ready.
— Reassignment on battery/sensor failure — automatic.

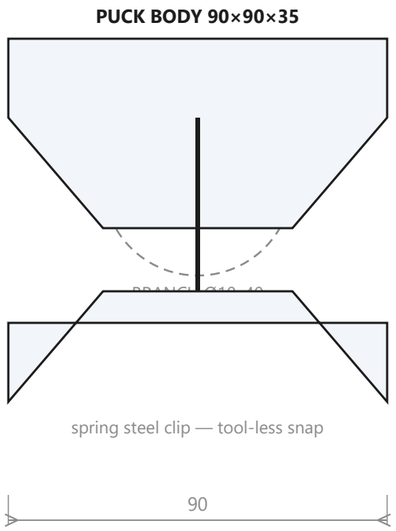
COST AT VOLUME (EST.)

VOLUME	SENSOR	RELAY	ALL VARIANTS BLENDED
1–10 units	$90–130 ea	$95–140 ea	≈$110 avg
100 units	$55–75 ea	$60–80 ea	≈$65 avg
1,000 units	$35–50 ea	$38–55 ea	≈$42 avg
10,000+	$20–30 ea	$22–32 ea	≈$25 avg — seed-drop economics

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UNITS **mm**

PERCH CLIP — SECTION



PERCH BEHAVIOR

— Clip grips branch/fence/rock — unit hangs vertically, solar faces up-ish.
— Wind-shake test: survives 60 km/h gusts on Ø25 mm branch.
— Perch sites chosen by mesh: fuel breaks, ridgelines, creek crossings.
— Docked variant (CHARGER-BEE) swaps clip for pogo alignment feet.

MANUFACTURING

— Shell/lid/bay: FDM print ASA or PC-ABS — or injection mold at volume.
— Clip: laser-cut 0.8 mm spring steel, powder-coated.
— Gasket: die-cut TPU sheet or printed-in-place.
— Carrier PCB: JLCPCB 4-layer ENIG — $15–30/pc at 10+.
— Assembly: hand-solderable — no BGA, no reflow oven needed.

ENCLOSURE STACK-UP

TOP LID — PC-ABS
2.0 mm — snap fit + gasket groove

SOLAR WINDOW — PC film
1.0 mm — 90% transmission, UV stable

MAIN SHELL — ASA
2.5 mm — gas-permeable vent zone

CARRIER PCB — FR4
1.6 mm — 38×38, ENIG finish

BATTERY BAY — PETG
3.0 mm — slide-in, JST access

Total Z ≈ 35 mm body + 10 mm clip = 45 mm field height.

SEALING & SERVICE

FEATURE	DETAIL
Gasket	TPU bead around lid — IP65 splash
Vent	ePTFE membrane — gas exchange, no water
Solar	under-window film — replaceable
Battery	slide tray — swap in field, no tools
PCB	2× M2 screws — hand driver only
Clip	spring steel — replaceable spare

DIMENSIONS

DIM	VALUE
Body	90 × 90 × 35 mm (L×W×H)
Clip span	Ø10–40 mm branch
Weight	~180 g with battery
Solar cell	62 × 62 mm active area
Carrier PCB	38 × 38 mm, 4-layer
Operating	-30..+60 °C (rev-A)

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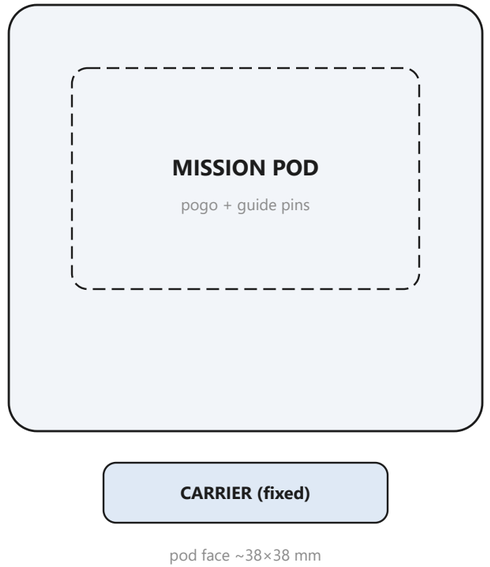
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UNITS **mm**

MISSION TAXONOMY

MISSION CLASS	EXAMPLE PAYLOADS	TRL / STATUS	BUYER
DETECT & MONITOR	ENS160+SHT41, thermal IR, GPS, mic	TRL 6–7 — prototype flies	fire/ag/municipal
ENVIRONMENTAL CLEANUP	microplastic skimmer, sheen recovery, invasive pull/sever tool, debris grab	TRL 2–3 — actuation R&D	watershed/industry
RESEARCH — REACH	gas sampler, water sampler, stereo imager, contact sensor	TRL 3–4 — payloads exist, miniaturize	universities/parks
TRANSPORT	payload ferry between Anchors	TRL 2 — concept	remote ops/industry

Carrier is FIXED: ESP32-S3 + SX1262 + solar + perch + mesh identity. Mission pods hot-swap on the top bay — same open-bay philosophy as the ArcticPhone. A bee re-tasks by swapping a pod, not buying a new bee.

POD MECHANICS (TOP VIEW)



MISSION PODS (TOP-BAY MODULES)

POD	HARDWARE DELTA	NOTES
SENSE-POD	ENS160 + SHT41 + mic + GPS	base — every swarm needs this
EYE-POD	AMG8833/MLX90640 thermal + lens	confirm fixes visually/thermally
CLEAN-POD	skimmer mesh / grabber / sever tool	actuation — rev-C, gated
LAB-POD	gas+water sampler, pH/conductivity	research missions — rev-C
RELAY-POD	8 dBi whip, sensors removed	range extension over ridges
CHARGE-POD	dock pads + beacon	Anchor perch — rebroadcast node

MISSION SEQUENCING

— Coordinator reads swarm capability flags + threat model.
— Season map: fire → fuel breaks; melt → river corridors; spills → water patrol; research → unreachable sites.
— Pods pre-staged at Anchor — swap takes <60 s, hand tools only.
— Failed pod = bee reverts to SENSE-POD and keeps watching.
— Every mission logged on mesh — provenance for regulators.

HONEST GATES

— Sensing swarm flies FIRST — cleanup/research are payload programs.
— Actuation payloads need their own safety review + pilot funding.
— No promise of cleanup before the sensing mechanism is proven.
— Transport is the furthest horizon — listed for platform completeness.

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SCALE **NTS**
UNITS **mm**

S · MITE
FINISH & MATERIALS — FINAL LOOK

AF-S-108
REV A · 2026-11-12

APPEARANCE REV A



S MITE — SENSOR-BEE
90 × 90 × 35 mm · ~180 g · forest camouflage

COLOR / FINISH OPTIONS

FOREST
default — blends in

SAGE
arid grassland

HI-VIS
training/retrieval

MATERIALS SPEC

COMPONENT	MATERIAL	FINISH	NOTE
Shell	ASA or PC-ABS	matte — forest tones	UV stable, -30 °C
Solar	poly cell + PC window	anti-reflective	replaceable film
Gasket	TPU bead	—	IP65 splash
Clip	spring steel 0.8 mm	olive drab powder-coat	field spare
Labels	laser-etched ID	—	swarm serial + role

DESIGN NOTES

— Forest green is deliberate — swarm should disappear into canopy.
— Hi-vis variant ONLY for bench testing and recovery ops.
— Solar window slightly recessed — survives branch strikes.
— Role variants share shell — only top insert + antenna differ.
— No logos on top face — solar area is sacred.

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UNITS **mm**